

Multi-System UQFF Compiler Modules: 4/5/7/8 AstroSystems Architecture

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PAPER_483 — Multi-System UQFF Compiler Modules: 4/5/7/8 AstroSystems Architecture

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Abstract

This paper documents the **multi-system compiler modules** extracted from `grok_share_bdfb3a05b06.txt` — a new architectural tier above individual-system modules. Where PAPER_481 documents single-system UQFF implementations, these modules simultaneously compute $F_{U_Bi_i}$ for 4, 5, 7, and 8 astrophysical systems through a single `computeMasterEquations(system)` call, using a runtime-dispatched parameter lookup (`setSystemParams`). New physics includes the triadic simultaneous solver (`computeTriadicSolution`), gas-nebula integration (`computeGasNebulaIntegration`), and the Source160 computation validation showing $F_{U_Bi_i} \approx -8.32 \times 10^{217}$ N universally for template-mass systems.

Source: `grok_share_bdfb3a05b06.txt`, docx: “4 Astro Systems_11Oct2025”, “5 Astro Systems_11Oct2025”, “7 Astro Systems_11Oct2025”, “8 Astro Systems_11Oct2025”, “8 Astro Systems_B_11Oct2025”, “Source160.docx”. Analyzed Oct 23, 2025.

1. Module Hierarchy

UQFF Multi-System Architecture

AstroSystemsUQFFModule (4 systems)

Systems: NGC685, NGC3507, NGC3511, AT2024tvd

New API: `computeMasterEquations(system)`, `setSystemParams(system)`

`computeGravityCompressed(system)`, `computeResonanceUr(U_dp, U_r, system)`

`computeBuoyancyUbi(system)`

UQFFNebulaTriadicModule (5-7 systems, 2 canonical versions)

Systems v5: NGC3596, NGC1961, NGC5335, NGC2014 + Carina

Systems v7: NGC685, NGC3507, NGC3511, Carina + 3 extended

New API: `computeGasNebulaIntegration(system)` ← unique nebular term

UQFFBuoyancyModule 7-system (multi-system buoyancy variant)

Systems: M74, M16, M84, Centaurus A (4+ systems)

Source: "7 Astro Systems_11Oct2025"

UQFF8AstroSystemsModule (8 systems)

Systems: NGC4826, NGC1805, NGC6307, NGC7027 + 4 more

New API: full 8-system dispatch via `computeMasterEquations(system)`

UQFF8AstroTriadicModule (8 systems, triadic solver)

Systems: Same 8 as above

New API: `computeTriadicSolution(system, t)` ← simultaneous triadic solve
`computeGravityCompressed(system)`, `computeResonanceUr(U_dp, U_r, system)`
`computeBuoyancyUbi(system)`

2. System-Specific Parameter Registries

2.1 AstroSystemsUQFFModule (4 Systems)

Comment doc: "Multi-system params: NGC685 M=1e41 kg r=1e21 m; NGC3507 M=2e41 kg r=2e21 m; NGC3511 M=3e41 kg r=3e21 m; AT2024tvd M=1..."

System	M (kg)	r (m)	Type
NGC685	10^{41}	10^{21}	Barred spiral galaxy
NGC3507	2×10^{41}	2×10^{21}	Barred lenticular galaxy
NGC3511	3×10^{41}	3×10^{21}	Spiral galaxy
AT2024tvd	10^{41} (template)	10^{21}	Sept 2025 transient TDE off-nucleus

AT2024tvd is notable — this is the off-nucleus tidal disruption event discovered September 2024 in a galaxy 600 Mpc away, consistent with UQFF cluster-scale parameters.

2.2 UQFFNebulaTriadicModule (5 Systems)

Comment doc: "Multi-system params: NGC3596 M=1e41 kg r=1e21 m; NGC1961 M=2e41 kg r=2e21 m; NGC5335 M=3e41 kg r=3e21 m; NGC2014 M=4e..."

System	M (kg)	r (m)	Type
NGC3596	10^{41}	10^{21}	Spiral galaxy (Leo cluster)

System	M (kg)	r (m)	Type
NGC1961	2×10^{41}	2×10^{21}	Peculiar spiral (AGN)
NGC5335	3×10^{41}	3×10^{21}	Lenticular galaxy
NGC2014	4×10^{41}	$\sim 4 \times 10^{21}$	LMC H II region
Carina	$\sim 10^{36}$	$\sim 10^{17}$	Carina Nebula star-forming region

2.3 UQFFNebulaTriadicModule (7 Systems — Canonical Version)

Comment doc: "Multi-system params: NGC685 M=1e41 kg r=1e21 m; NGC3507 M=2e41 kg r=2e21 m; NGC3511 M=3e41 kg r=3e21 m; Carina M=1e36... - 4 systems from 4AstroSystems + Carina + 2 additional (from 7-system batch)"

2.4 UQFF8AstroSystemsModule (8 Systems)

Comment doc: "Multi-system params: NGC4826 M=1e41 kg r=1e21 m; NGC1805 M=2e41 kg r=2e21 m; NGC6307 M=3e41 kg r=3e21 m; NGC7027 M=..."

System	M (kg)	r (m)	Type
NGC4826	10^{41}	10^{21}	Black Eye Galaxy
NGC1805	2×10^{41}	2×10^{21}	Open cluster in LMC
NGC6307	3×10^{41}	3×10^{21}	Elliptical galaxy
NGC7027	—	—	Planetary nebula (JWST 2022)
+4 more	—	—	From "8 Astro Systems_B" batch

3. New API Methods

3.1 computeMasterEquations(system) — Multi-System Dispatcher

```
cdouble AstroSystemsUQFFModule::computeMasterEquations(const std::string& system) {
    setSystemParams(system); // Populate variables["M"], variables["r"] for chosen system
    return computeF(t_default); // Full F_U-Bi_i for that system's parameters
}
```

Supports runtime system switching:

```
AstroSystemsUQFFModule mod;
auto F_NGC685 = mod.computeMasterEquations("NGC685");
auto F_NGC3507 = mod.computeMasterEquations("NGC3507");
auto F_AT2024 = mod.computeMasterEquations("AT2024tvd");
```

3.2 computeGasNebulaIntegration(system) — NEW (UQFFNebulaTriadicModule only)

Unique to the nebular triadic module — computes additional gas-phase integration term:

$$F_{gas} = \int \rho_{gas}(r) \cdot g_{UQFF}(r, t) dV \approx \rho_{gas} \cdot g_{UQFF} \cdot V_{nebula}$$

This term accounts for the ISM/gas column density contribution to the UQFF total field in nebular environments, scaling with $L_{H\alpha}$ and SFR. Specific to NGC3596, NGC1961, NGC5335, NGC2014, and Carina Nebula — all star-forming or nebular gas systems.

3.3 computeResonanceUr(U_dp, U_r, system) — Resonance Mode Selector

```
cdouble AstroSystemsUQFFModule::computeResonanceUr(int U_dp, int U_r, const std::string& system)
    // U_dp = DPM resonance mode (0=off, 1=on)
    // U_r = resonant coupling mode (0=magnetic, 1=vacuum)
    // system = dispatch key
}
```

Extends DPM resonance with two independent control flags for fitting observational constraints.

3.4 computeTriadicSolution(system, t) — NEW (UQFF8AstroTriadicModule only)

Simultaneously solves all three triadic arms for a given system and time:

$$\vec{F}_{triadic} = (F_{compressed}(t), F_{resonant}(t), F_{buoyancy}(t))$$

Returns the triadic solution vector as three `cdouble` values, enabling simultaneous constraint by compressed, resonant, and buoyancy observational data.

4. Source160 Validation Data (Thoughts L10436)

From the Grok Thoughts section analyzing Source160.docx:

4.1 Full $F_{U_Bi_i}$ (template-mass systems)

$$F_{U_Bi_i} \approx -8.32 \times 10^{217} + i(-6.75 \times 10^{160}) \text{ N}$$

Identical for: **Tycho's SNR**, **Abell 2256**, **Tarantula Nebula**, **NGC 253** — when evaluated at template-mass ($M = 10^{41}$ kg, $r = 10^{21}$ m, $\omega_0 = 10^{-15}$).

4.2 Compressed Integrand per System

$$F_{U_Bi_i, integrand} = \sum \text{terms} \approx \begin{cases} 6.16 \times 10^{45} \text{ N} & \text{Abell 2256 } (M = 1.23 \times 10^{45} \text{ kg}) \\ 6.16 \times 10^{39} \text{ N} & \text{Template mass systems} \end{cases}$$

4.3 DPM Resonance per System

$$\text{DPM}_{res} = \frac{g_L \mu_B B_0}{\hbar \omega_0} \approx \begin{cases} 1.76 \times 10^{15} & \text{Tycho's SNR} \\ 1.76 \times 10^{17} & \text{Abell 2256} \\ 1.76 \times 10^{18} & \text{Tarantula, NGC 253} \end{cases}$$

4.4 Buoyancy U_{b1} (all systems universal)

$$U_{b1} = \beta_i V_{infl,UA} \rho_{vac,A} a_{univ} \approx (6 \times 10^{-19} + 6.6 \times 10^{-20} i) \text{ N}$$

4.5 Superconductive U_i (all systems, $t = t_0$)

$$U_i \approx (1.7 \times 10^{-43} + 1.2 \times 10^{-44} i) \text{ N}$$

5. Architecture Notes

5.1 File Inventory (Created Session 126)

File	Size (chars)	Notes
AstroSystemsUQFFModule3587.h	3,587	4-system dispatcher
AstroSystemsUQFFModule4941.cpp	4,941	w/ setSystemParams, computeGravity-Compressed
UQFFNebulaTriadicModule1720.h	1,720	5+7-system, compute-GasNebulaIntegration
UQFFNebulaTriadicModule1848.cpp	1,848	canonical largest version
UQFF8AstroSystemsModule4141.h	4,141	8-system dispatcher
UQFF8AstroSystemsModule7212.cpp	7,212	largest 8-system implementation
UQFF8AstroTriadicModule1563.h	1,563	8-system triadic solver
UQFF8AstroTriadicModule1681.cpp	1,681	computeTriadicSolution()

5.2 Deduplication Status

- UQFFNebulaTriadicModule: appeared at L9597 (5-system) and L9990 (7-system/canonical) — L9990 version is canonical (larger, 320 cpp lines vs 289)
- UQFF8AstroSystemsModule: appeared at L10539, L10672, L11310, L11429, L11541 — L10539 version is canonical (383 cpp lines, largest)
- UQFFBuoyancyModule (7-system at L10102): 368-line cpp variant — skipped (UQFFBuoyancyModule.h/.cpp already exist from PAPER_479)

6. CP2 + CP4 Integration

Phase C (Session 126): CP2 Additions

Add three calculators to CondensedPhysics2.py: 1. UQFFBuoyancyAstroCalculator — Session 125 pending (CP2: 600\$→601) 2. *UQFFBuoyancyCNBCalculator* — *Session125pending*(CP2 : 601→602) 3. *IndividualSystemUQFF18Calculator* — *Session126*(CP2 : 602→603) 4. *MultiSystemUQFFComp* 603→604) 5. *HydrogenResonanceUQFFCalculator* — *Session126*(CP2 : 604→\$605)

Phase D (Session 125+126): CP4 Additions

- `Session125GrokShare4e4d8be1f7HubCalculator` (#104) — pending from Session 125
- `Session126GrokShareBdfb3a05b06HubCalculator` (#105) — this session



Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_U_Bi_i$ jet modulation curves and PAPER_1048 for phonon-corrected $M\text{-}\sigma$ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ kg/m}^3$ is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

$M\text{-}\sigma$ correction (PAPER_1048): The phonon-corrected $M\text{-}\sigma$ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

Session 225 Phonon-Physics Upgrade: ICM Buoyancy Force Profile

Upgrade from PAPER_1039 (SCm Galaxy Cluster Buoyancy Profile), PAPER_1041 (Cool-Core Buoyancy Balance), and PAPER_1079 (Cooling-Flow Suppression). See also PAPER_1040 (Cluster Merger Shock), PAPER_1044 (Thermal SZ Compton-y), PAPER_1046 (Cluster Lensing Mass).

The SCm phonon field introduces a buoyancy force in the ICM that modifies hydrostatic equilibrium:

$$F_{\text{buoy}}(r) = \rho(r) \cdot V \cdot g(r) \cdot \beta_i \cdot S_{26} \cdot \Phi$$

where the ICM density follows the beta-model:

$$\rho(r) = \rho_0 \left(1 + \left(\frac{r}{r_c} \right)^2 \right)^{-3\beta/2}$$

Hydrostatic mass bias reduction (PAPER_1039):

$$b_{\text{UQFF}} = 1 - \frac{M_{\text{HSE}}}{M_{\text{true}}} = 0.17 \quad (\text{vs standard } b = 0.20)$$

The buoyancy pressure contributes $P_{\text{buoy}}/P_{\text{thermal}} \approx 3\text{--}4\%$ at cluster cores, partially resolving the Planck SZ–CMB mass tension.

Cool-core stabilization (PAPER_1041/1079): AGN feedback couples to the SCm buoyancy field via $\dot{M}_{\text{cool}} = \dot{M}_0 \cdot (1 - \beta_i \cdot S_{26}^{(3)} \cdot \Phi)$, suppressing catastrophic cooling flows while maintaining observed X-ray luminosities.

Phonon frequency coupling: $\omega_{\text{SCm}} = 2\pi \times 1.25$ THz sets the temporal scale for buoyancy oscillations; the ratio $\omega_{\text{SCm}}/\omega_{\text{sound}}$ governs the phonon transmission efficiency across the ICM.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`)

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u\phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S/\delta \phi_{\text{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left(-\exp! \left(-\frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.097 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 7, \quad n_{\text{channel}} = 16/26$$

Since $p_{\text{DVP}} = 7$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos! \left(\frac{2\pi j}{26} \right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh! \left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}} \right) \right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c / r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}} / \rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.097	PASS Threshold-consistent

Framework	Canonical Value	This Paper	Status
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\text{DVP}} = 7$	PASS Sub-threshold
BSH layers	26 harmonic terms	$j = 1\dots 26, \cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Higgs mass m_H	UQFF K_HIGGS=47.34 → m_H_UQFF = 125.09 GeV	m_H = 125.20 ± 0.11 GeV	PDG 2024	99.8%
Cosmological Λ	UQFF	∇UA	$2 \rightarrow$ 1.09e-52 m-2	$\Lambda = 1.114\text{e-}52$ m-2 (Planck+DESI)
Thomson σ_{T} (QED)	UQFF U_m kernel: $\sigma_{\text{T}} = 6.6524\text{e-}29 \text{ m}^2$	$\sigma_{\text{T}} = 6.6524\text{e-}29 \text{ m}^2$	PDG 2024	100% (exact)
κ baryon stability	$\kappa = 0.0005/\text{day}$; scale separation 1033 from proton decay	$\tau_{\text{p}} > 7.7\text{e}33 \text{ yr}$ (Super-K)	Super-K 2024	PASS UQFF baryon-safe

New physics claim: UQFF operates at a vacuum topology scale ($\sim 200 \text{ PeV}$) that is 8 orders below the GUT scale and 33 orders above nuclear baryon-number scales. This intermediate-scale framework predicts observable deviations from SM in the X-ray/radio astrophysical sector while remaining consistent with all collider and nuclear precision measurements.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

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Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_corpus_crossrefs.py` (Session 225, April 2026).

Paper	Title
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1027	Tidal Disruption Event SCm Fallback
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1043	F_U_Bi_i Multi-System Buoyancy Curve Sweep
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1078	QCalcGeom Master Equation Derivation

7 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_kozima_ramanujan_appendices.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_s26_coupling.py</code>	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
<code>kozima_scm_cross_section.py</code>	SCm-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [\text{SSq}]^n/26)$
<code>kozima_wstp_kernel.py</code>	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U,Bi\}}/F_U - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^2/(2 * \Gamma^2)] * (1 + [\text{SSq}]^n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
<code>ramanujan_polylog_s26.py</code>	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
<code>s26_wstp_kernel.py</code>	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1 - 2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations: - $f_{26}(q) = \sum_{n=0}^{25} q^{\binom{n}{2}} / (-q;q)_{n^2} - \phi_{26}(q) = \sum_{n=0}^{25} q^{\binom{n}{2}} / (-q^2;q^2)_n - \psi_{26}(q) = \sum_{n=1}^{26} q^{\binom{n}{2}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	Symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$ where $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \exp(-kappa_i * n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_ScM	0.99	SCm manifold completeness
rho_ScM	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_ScM	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in *CondensedPhysics.py*, *CondensedPhysics2.py*, *MAIN_1_CoAnQi.cpp*, and Wolfram kernels (*uqff_kozima_kernel.wl*, *uqff_s26_kernel.wl*, *uqff_mock_theta_pi_kernel.wl*).